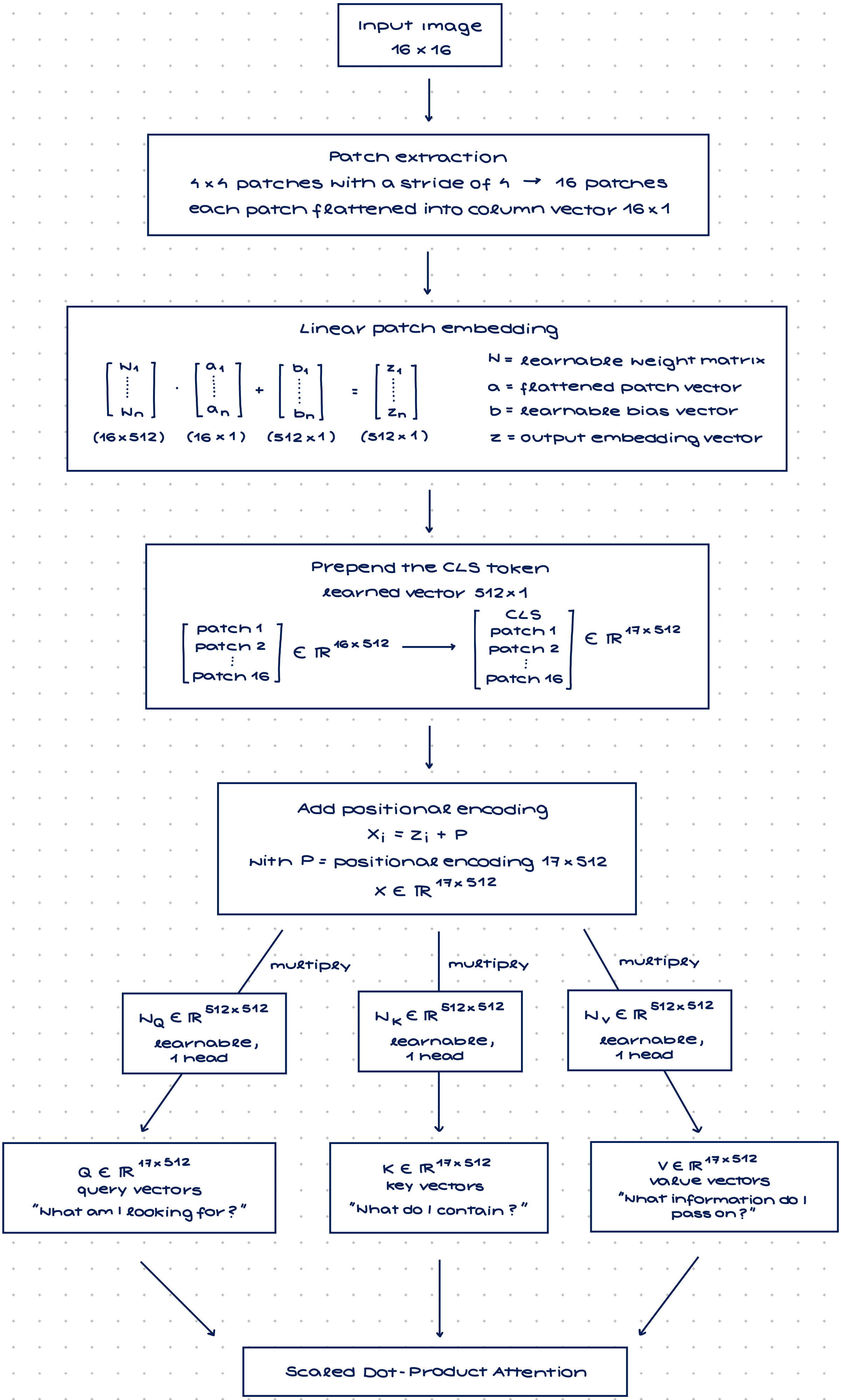
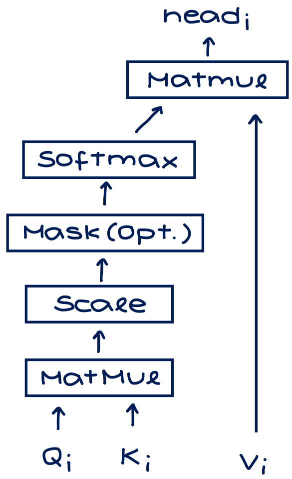
Novo Nordisk Project - Week 3

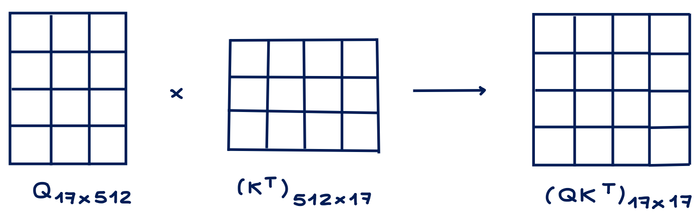
a. Submit a one-page overview of MHA mechanism



Scaled Dot-Product Attention

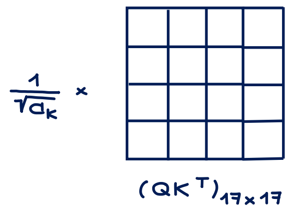


Step 1: Matrix Multiplication



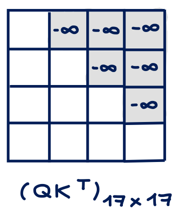
Step 2: Scale

→ variance normalisation



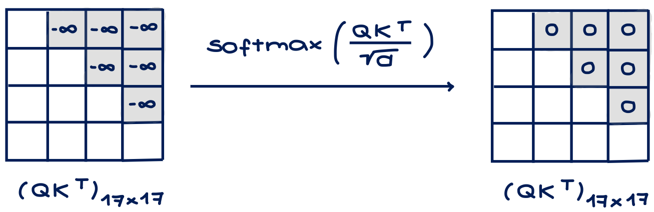
with $d_k = 512$

Step 3: Masking (optional)



! ViT uses an encoder, no mask applied

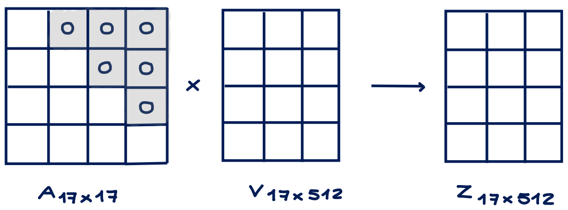
Step 4: Softmax



= how much attention token i pays to every other token

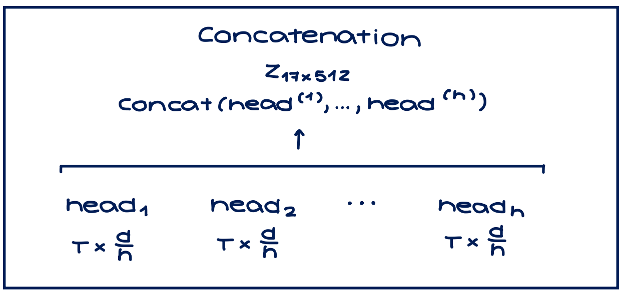
each row is an attention distribution for one token

Step 5: Matrix Multiplication



$A_{17\times17}$ = attention weights

$Z_{17\times512}$ = context matrix

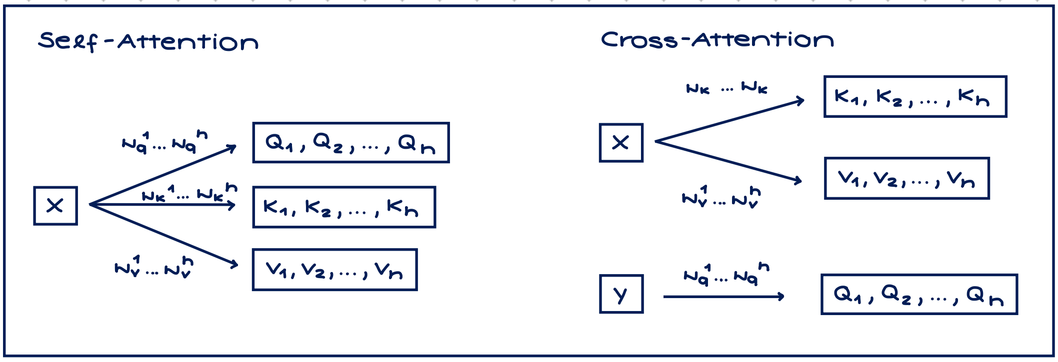


Linear Transformation

$\text{MultiHead}(Q, K, V) = Z \times W_{512\times512}$

$Z_{17\times512} = \text{concat}(head^{(1)}, \dots, head^{(h)})$

$W_{512\times512}$ = learnable weight matrix



What is a CLS token (class token) is a learnable vector that is added at the beginning of the input sequence to collect and represent information from all other tokens → final summary for classification

Novo Nordisk Project - Week 5

a. Submit a one-page overview of the entire transformer encoder

Input image
16 x 16

↓

Patch extraction
4 x 4

↓

Linear patch embedding

↓

Prepend the CLS token

↓

Add positional encoding
$x_i = z_i + P$

multiply | multiply | multiply

$W_Q \in \mathbb{R}^{512 \times 512}$
learnable,
1 head

$W_K \in \mathbb{R}^{512 \times 512}$
learnable,
1 head

$W_V \in \mathbb{R}^{512 \times 512}$
learnable,
1 head

↓

$Q \in \mathbb{R}^{17 \times 512}$
query vectors
"What am I looking for?"

$K \in \mathbb{R}^{17 \times 512}$
key vectors
"What do I contain?"

$V \in \mathbb{R}^{17 \times 512}$
value vectors
"What information do I pass on?"

↓

Scaled Dot-Product Attention

↓

Concatenation
$Z_{17 \times 512}$
$\text{concat}(\text{head}^{(1)}, \dots, \text{head}^{(h)})$

↓

Linear Transformation
$\text{MultiHead}(Q, K, V) = Z \times W_{512 \times 512}$

$Z_{17 \times 512} = \text{concat}(\text{head}^{(1)}, \dots, \text{head}^{(h)})$

$W_{512 \times 512}$ = learnable weight matrix

↓

Residual Connection
each token keeps their original representation and adds the new information

$R = X^{17 \times 512} + Y^{17 \times 512}$

Layer Normalization
applied to each token independently

Linear Expansion

$$\underset{(512\times 2048)}{\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}} \cdot \underset{(17\times 512)}{\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}} + \underset{(17\times 2078)}{\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}} = \underset{(17\times 2048)}{\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}}$$

W = learnable weight matrix
a = normalized token vector
b = learnable bias vector
V = output vector

GELU
Gaussian Error Linear Unit

$$\mathrm{GELU}(x) \approx x\,\Phi(x)$$

$\Phi(x) \triangleq$ cumulative distribution function

Linear Expansion

$$\underset{(2048\times 512)}{\begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}} \cdot \underset{(17\times 2048)}{\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}} + \underset{(17\times 512)}{\begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}} = \underset{(17\times 512)}{\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}}$$

W = learnable weight matrix
v = normalized token vector
b = learnable bias vector
y = output vector

Residual Connection
each token keeps their original representation and adds the new information

$$R = X^{17\times 512} + Y^{17\times 512}$$

CLS token extraction

Layer Normalization

Linear Layer

$$T = CLS^{1\times 512} \cdot W^{512\times 2}$$

$$T \in \mathbb{R}^{2\times 1}$$

Softmax
converts values into probability distribution

Classification layer

$$Y = \mathrm{argmax}(X) \in [0,1]$$